

Quantum Annealing for Boundary Point Detection in Binary Classification

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Abstract

This report presents a novel approach to binary classification by amalgamating techniques from computational and algebraic geometry with quantum annealing, traditionally disparate fields in computer science. Our methodology begins with the formation of a nearest neighbor graph of the dataset, followed by framing a graph problem aimed at identifying points residing on the classification boundary, referred to as Models of Classes Distribution (MCD) points. These MCD points serve as crucial indicators of class distribution patterns within the dataset. To implement this methodology, we leverage quantum annealing to efficiently solve the formulated IP problem using Graver Basis. Subsequently, an Ising model is employed to make predictions on unseen test data based on the distribution insights gleaned from the MCD points. It is important to note that our primary objective is not to outperform classical state-of-the-art classifiers in terms of computational efficiency or predictive accuracy. Through experimental results, we demonstrate the viability and potential insights gained from this unconventional fusion of computational and algebraic geometry with quantum annealing.

1 Introduction

Quantum Machine Learning, a burgeoning research area at the intersection of quantum computing and machine learning, has gained prominence owing to the inherently probabilistic nature of quantum systems, heralding a paradigm shift in classification methodologies. Harnessing the distinctive features of quantum systems, our aim is to pioneer a novel approach to binary classification.

Our approach involves modeling the classes of data points as spins, enabling the construction of an Ising model for making predictions on test data. Leveraging spatial information in the dataset to model spin coupling, we employ Quantum Annealing on D-Wave’s Quantum Processing Units (QPUs) [6] to solve the Ising model. However, as the size of the training dataset increases, so does the number of spins required, leading to heightened model complexity. This, in turn, results in an increased average chain length embedded on the QPU.

The D-Wave QPU operates on a grid of qubits and couplers, with the Advantage systems implementing the Pegasus Topology for qubit interconnection. Figure 1 illustrates an example embedding on the Pegasus architecture, where nodes of the same color chained together correspond to a single logical unit and their connections form chains. Each qubit within a chain is mathematically constrained to have the same value (-1 or 1). To ensure the reliability of results, an optimal chain strength value must be set, allowing the chain to resist unnecessary breaks. However, with an escalating average chain length, determining the correct chain strength becomes challenging, leading to unreliable predictions.

In light of these challenges, there arises a need to reduce the number of spins, hence the number of training data-points while maintaining the integrity of the model.

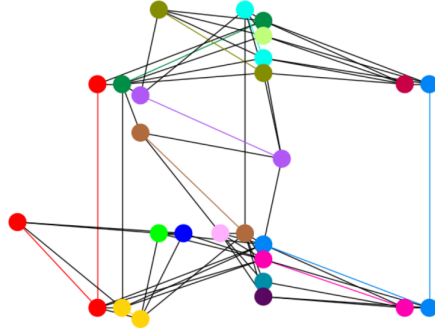


Figure 1: Example Problem Embedding on Pegasus Architecture in DWave-Advantage Systems [6].

Addressing the challenge of large training datasets draws parallels with longstanding issues encountered in Support Vector Machines (SVM). Historically, solutions have gravitated toward two categories: (1) modifying algorithms for scalability to large datasets and (2) reducing the dataset size by retaining representative points that encapsulate the classes distribution, termed Models of Classes Distribution (MCD) points [1], Figure 2 highlights the MCD points for a sample dataset. Choosing the latter path, we delve into an intriguing subproblem, worthy of dedicated study for its potential generalization and applications in graph and network theory, presenting an opportunity for unique investigation.

Prior approaches in the second category include leveraging Minimum En-

closing Ball (MEB) to identify support vectors constituting MCD points [1] and employing DBSCAN in conjunction with the quantum algorithm Deutsch-Jozsa for support vector selection [2].

Our proposed methodology for MCD point identification involves constructing a nearest neighbor graph of the dataset. We frame the challenge of identifying MCD points as an Integer Linear Programming model and employ the innovative GAMA approach for its solution [3]. Leveraging the properties of slack variables used in modifying the constraints, we successfully pinpoint the MCD points. Subsequently, these points serve as the foundation for constructing an Ising Prediction model, allowing accurate predictions on the test dataset. This unique approach not only addresses the challenge of large training datasets but also sheds light on the potential of Integer Linear Programming and geometric algorithms in the context of quantum machine learning.

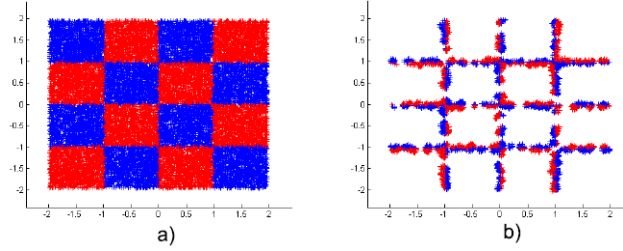


Figure 2: Figure showcasing the points considered as MCD points [1].

2 Literature Review

2.1 Delaunay Triangulation [5]

In our approach, the construction of a graph connecting the points is crucial, and it should reflect the spatial relationships between the points, at least to some approximation. Various types of graphs and approximation methods exist, each with its own merits and drawbacks. For our purposes, we opted to use a Delaunay triangulation graph.

A triangulation of a finite point set $\mathbf{P} \subset \mathbb{R}^2$ is termed a Delaunay triangulation if the circumcircle of every triangle is empty, meaning there is no point from $\mathbf{P}$ in its interior. Our choice of the Delaunay triangulation is motivated by the desire to showcase our algorithm in two dimensions, facilitating better visualization. Additionally, Delaunay triangulation is more efficient in 2D, connecting each point to its natural neighbors.

While the time complexity of Delaunay triangulation increases with the dimension of the problem, various approximation methods exist to mitigate this issue. One such approach is the Monte Carlo ray tracing method.

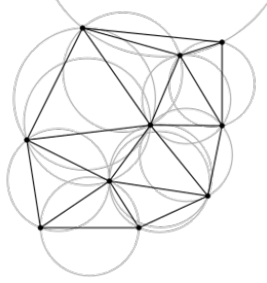


Figure 3: Figure shows the Delaunay triangulations with circumcircles. Credits:Wikipedia

2.2 Graver Augmented Multiseed Algorithm [3]

2.2.1 Graver Basis

In combinatorics and integer programming, the Graver basis denoted $\mathcal{G}(A)$ for an integer matrix A , is a set of conformal minimal elements in the lattice $\mathcal{L}^* = \{\mathbf{x} \mid A\mathbf{x} = \mathbf{0}, \mathbf{x} \in \mathbb{Z}^n, A \in \mathbb{Z}^{m \times n}\} \setminus \{\mathbf{0}\}$. Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. We say $\mathbf{x}$ is conformal to $\mathbf{y}$, written $\mathbf{x} \sqsubseteq \mathbf{y}$, when $x_i y_i \geq 0$ ($\mathbf{x}$ and $\mathbf{y}$ lie on the same orthant) and $|x_i| \leq |y_i|$ for $i=1, \dots, n$.

A set $S \subseteq \mathbb{Z}^n$ is called a test set or optimality certificate if for every non-optimal but feasible solution x_0 there exists $t \in S$ and $\lambda \in \mathbb{Z}_+$ such that $x_0 + \lambda t$ is feasible and $f(x_0 + \lambda t) < f(x_0)$. The vector t (or λt) is called the improving or augmenting direction. If the optimality certificate is given, any initial feasible solution x_0 can be augmented to the optimal solution. If S is finite, one can enumerate overall $t \in S$ and check if it is augmenting(improving). If S is not practically finite, or if all elements $t \in S$ are not available in advance, it is still practically enough to find a subset of S that is feasible and augmenting. Graver Basis forms one such test set.

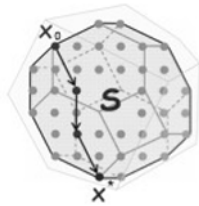


Figure 4: Augmenting from initial to optimal solution. [4]

2.2.2 GAMA Approach

We need to solve $\mathbf{Ax} = \mathbf{0}, \mathbf{x} \in \mathbb{Z}^n, \mathbf{A} \in \mathbb{Z}^{mn}$, which is computationally hard, in a bounded region. Instead, we look to find a large sampling of the non-zero elements of the kernel. We will map the Kernel problem into a QUBO. Each call of a quantum-annealing-based QUBO solver can run through multiple reads and extract several (unique) degenerate ground states. We expect to be able to compute a large sampling of the kernel of $\mathbf{A}$ in one or more calls (of 1000 reads each). To do this, we use quadratic unconstrained integer optimization (QUIO):

$$\min \mathbf{x}^T \mathbf{Q_I} \mathbf{x}, \quad \mathbf{Q_I} = \mathbf{A}^T \mathbf{A}, \quad \mathbf{x} \in \mathbb{Z}^n$$

$$\mathbf{x}^T = [x_1 \quad x_2 \quad \dots \quad x_i \quad \dots \quad x_n], \quad x_i \in \mathbb{Z}$$

Quantum Annealers solves the quadratic unconstrained binary problems, So we have to encode the integer variables into binary variables using suitable mapping. To map an integer variable to a binary, while keeping the problem quadratic, we use a linear mapping: $x_i = \mathbf{e}_i^T X_i$ where $\mathbf{e}_i^T = [e_i^1 \quad e_i^2 \quad \dots \quad e_i^{k_i}]$ $i \in \mathbb{Z}_+^{k_i}$ is the linear encoding vector and $X_i^T = [X_{i,1} \quad X_{i,2} \quad \dots \quad X_{i,k_i}] \in 0, 1^{k_i}$ is the encoded binary vector that encodes integer value x_i . The k_i is the length (number of bits) of encoding vector e_i , and depends on the linear encoding scheme that is in use. By choosing the same encoding for size for all integer variables $\mathbf{e}_i = \mathbf{e}_g, k_i = k_g \forall i = 1, 2, \dots, n$ with encoding matrix $(n \times nk_g)$ $\mathbf{E}_g = \mathbf{e}_g \otimes \mathbf{I}_n$ where $\otimes$ is the Kronecker product and $\mathbf{I}_n$ is the identity matrix of size n . Therefore the quadratic unconstrained binary optimization formulation simplifies to,

$$\min \mathbf{X}^T \mathbf{Q_g} \mathbf{X}, \quad \mathbf{Q_g} = \mathbf{E}_g^T \mathbf{Q_I} \mathbf{E}_g + \text{diag}(2\mathbf{L}_g^T \mathbf{Q_I} \mathbf{E}_g)$$

$$\mathbf{X} \in \{0, 1\}^{nk} \quad \mathbf{Q_I} = \mathbf{A}^T \mathbf{A}$$

Where, $\mathbf{L}$ is a $(n \times 1)$ vector containing the integer variable's lower bounds. After acquiring the kernel set K_i in call iteration i by randomly permuting the columns of $\mathbf{A}$ as referred in the paper [3], by sifting the $\sqsubseteq$ minimal elements of the kernel into Truncated Graver basis $\mathcal{G}(A)$, with which we start augmentation procedure on the set of initial feasible solution we obtain using the procedure described below.

"We can derive a feasible solution by solving the linear system $\mathbf{Ax} = \mathbf{b}$. Analogous to the methodology used for solving the Kernel elements, we will employ the Quadratic Unconstrained Binary Optimization (QUBO) approach to address this problem."

$$\min \mathbf{X}^T \mathbf{Q_f} \mathbf{X}, \quad \mathbf{Q_f} = \mathbf{E_f}^T \mathbf{Q_I} \mathbf{E_f} + 2\text{diag}(\mathbf{L_f}^T \mathbf{Q_I} - \mathbf{b}^T \mathbf{A}) \mathbf{E_f}$$

Where $\mathbf{E_f} = \mathbf{e_f} \otimes \mathbf{I}_n$, $\mathbf{e_f}$ being the encoding vector used in getting the initial seeds for augmentation. After obtaining the feasible solutions, we start from a feasible solution then augment using the Truncated Graver basis set. The path

between the initial feasible solution and the unknown optimal solution falls into a specific orthant based on their positions. Deriving the Graver basis inside this orthant quickly leads to reaching the optimal solution. Thus, starting with multiple feasible solutions and performing augmentation on them in parallel increases the likelihood of reaching the optimum value faster during the iterative process of building the Graver basis.

3 Methodology

3.1 Training Phase

3.1.1 Integer Linear Programming Formulation

We formulate an Integer Linear Programming (ILP) problem on the graph $G(V, E)$ derived from the dataset, where each $v_i \in V$ is assigned the value y_i (representing the class label of the point). The objective is to partition G into three subsets, namely S_0 , S_1 , and S_2 , such that nodes in S_0 do not share an edge with nodes in S_2 , and vice versa. The subset S_1 is strategically placed to act as a separator between the two partitions. By enforcing S_0 to predominantly include points belonging to class +1 and S_2 to include points of class -1, the objective function aids in identifying points forming the Models of Classes Distribution (MCD) within the S_1 subset.

We introduce an integer variable x_i for each $v_i \in V$ representing the subset to which the point belongs, where $x_i \in \{0, 1, 2\}$. The ILP formulation is given by:

$$\begin{aligned} \max_{\mathbf{x}} \quad & \frac{1}{2} \sum_i (x_i(1 - x_i) + (1 - x_i)(2 - x_i)) y_i \\ \text{s.t.} \quad & -1 \leq x_i - x_j \leq 1 \quad \text{for } (v_i, v_j) \in E \end{aligned}$$

where $\mathbf{x} = [x_1 \ x_2 \ \dots \ x_n]^T$.

Figure 5 illustrates this objective using an example dataset. It is evident that one subset covers most of the red points, the other most of the blue points, and the S_1 subset effectively separates them, containing the points constituting the MCD. While these points are crucial for understanding class distribution, it is recognized that additional refinement may be required. Figure 6 demonstrates how the constraints operate, ensuring that points sharing an edge are prohibited from assuming values (0,2) and (2,0), thereby enforcing the separation of S_0 and S_2 subsets.

3.1.2 Modification of Constraints

To incorporate the GAMA approach, it is necessary to convert the original inequality constraints into equality constraints. Additionally, modifying these

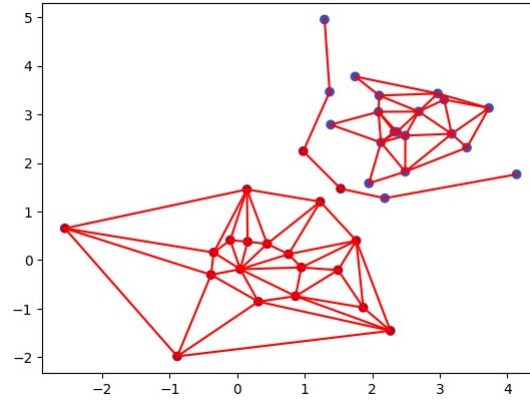


Figure 5: Example figure illustrating the three subsets for a sample dataset.

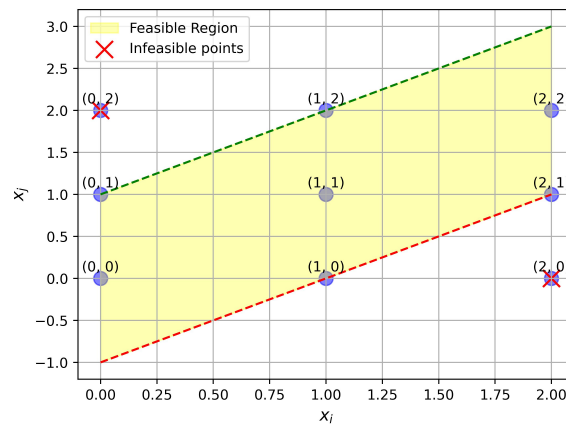


Figure 6: Plot illustrating the constraints.

constraints can enhance our ability to capture more points constituting the Models of Classes Distribution (MCD). The initial constraint, $-1 \leq x_i - x_j \leq 1$ for each edge $e \in E$, is transformed into an equality constraint by introducing a slack variable $s_e \in \{0, 1, 2\}$, resulting in the new constraint $x_i - x_j + s_e = 1$. Table 1 illustrates how this modified equality constraint enforces the original inequality constraint.

Table 1: Feasibility table for modified Constraint

x_i	x_j	s_e	Feasibility
0	0	1	Feasible
0	1	2	Feasible
0	2	3	Infeasible
1	0	0	Feasible
1	1	1	Feasible
1	2	2	Feasible
2	0	-1	Infeasible
2	1	0	Feasible
2	2	1	Feasible

We observe that when a slack variable s_e takes values 0 or 2, the corresponding edge represents a connection between subsets S_0 and S_1 or between S_1 and S_2 . Leveraging this insight allows us to identify additional points within the Models of Classes Distribution (MCD) that also encompass the points initially residing in the subset S_1 . This is achieved by considering the points $(\mathbf{x}_i, \mathbf{x}_j)$ forming these edges e whose corresponding slack variable s_e takes a value of 0 or 2, subject to the condition $y_i y_j \leq -1$.

3.1.3 Implementing GAMA

For obtaining the Graver basis, we employed the encoding vector $\mathbf{e}_g = [-1 \ 1]^T$ with the lower bound vector L_g being a zero vector and permuted the columns of matrix A randomly in each iteration. Subsequently, for obtaining the initial feasible set, to confine the elements of feasible solutions to the set $\{0, 1, 2\}$, we utilized the encoding vector $\mathbf{e}_f = [1 \ 1]^T$ again with lower bound vector L_f being zero vector. However, this encoding led to a significant occurrence of 1s in feasible solutions, creating compatibility issues with the truncated Graver basis set.

To address this, we introduced a penalty term $Q_p = \mathbf{E}_f^T \mathbf{E}_f - 2\text{diag}(\mathbf{1}^T \mathbf{E}_f)$ to our QUBO matrix. Here, $\mathbf{1}^T$ is a row vector of 1s, and $\mathbf{E}_f$ represents the encoding matrix used in obtaining the initial feasible solutions. The final QUBO matrix for feasible solutions is adjusted as $\mathbf{Q}_f' = \mathbf{Q}_f - \lambda \mathbf{Q}_p$, where λ is the penalty parameter.

3.2 Prediction Phase

In the prediction phase, we employ an Ising model to make predictions based on the Models of Class Distribution (MCD) obtained during the training phase and a test dataset D_{test} . The Ising prediction model is designed to incorporate information from the labels of the truncated training data containing only the MCD points(D_b), ensuring the integration of distinctive characteristics from each class. Furthermore, the model considers the relative positions between data points through coupling, capturing spatial relationships within the dataset.

3.2.1 Datasets

We have a training dataset containing only the MCD points D_b consisting of labeled instances $\{\mathbf{x}_n, y_n\}_{n=1}^{N_b}$ and a test dataset D_{test} containing instances $\{\mathbf{x}_n, y_n\}_{n=1}^{N_{test}}$

3.2.2 Ising Model Definition

The total number of spins is $N_{spins} = N_b + N_{test}$, and the Hamiltonian is defined as

$$H = - \sum_i h_i s_i - \sum_{i < j} J_{ij} s_i s_j$$

. Here, h_i represents the magnetic field created by the boundary points, and we set,

$$h_i = \begin{cases} \alpha y_i, & \text{for } \mathbf{x}_i \in D_b \\ 0, & \text{otherwise} \end{cases}$$

The parameter α is chosen to be large enough to ensure the spins of the training data remain unchanged during annealing.

3.2.3 Coupling Coefficients

The coupling coefficients J_{ij} are determined based on the training and test datasets. Specifically,

$$J_{ij} = \begin{cases} K_{\beta_1}(\mathbf{x}_i, \mathbf{x}_j), & \text{if } \mathbf{x}_i \text{ \& } \mathbf{x}_j \in D_{test} \\ K_{\beta_2}(\mathbf{x}_i, \mathbf{x}_j), & \text{if } \mathbf{x}_i \in D_{test}, \mathbf{x}_j \in D_b \\ 0, & \text{otherwise} \end{cases}$$

Here,

$$K_{\beta} = \frac{\beta}{\|\mathbf{x}_i - \mathbf{x}_j\|_2^2}$$

, The norm used being L_2 norm, the points that are closer will have stronger coupling than the points that are farther apart.

3.2.4 Varying Parameters

By varying the values of parameters β_1 and β_2 , we obtain new decision boundaries. The annealing process adapts the Ising model to the dataset's spatial characteristics, providing a flexible mechanism to explore different configurations.

The Ising model, characterized by its Hamiltonian and coupling coefficients, enables us to predict class labels for the test dataset based on the learned patterns from the Models of Class Distribution. This adaptable approach allows for the exploration of diverse decision boundaries, enhancing the model's predictive capabilities.

4 Results

We utilized a synthetic dataset comprising two classes, each containing 20 points in 2D space to showcase our approach. Class 1 points were generated from a standard normal distribution, while Class 2 points were also generated from the same distribution but with an additional constant offset of 3 in both axes. This offset effectively shifted the distribution of points for Class 2 to the right and above the distribution of points for Class 1.

Figure 7 presents the dataset along with its Delaunay 1-simplices graph.

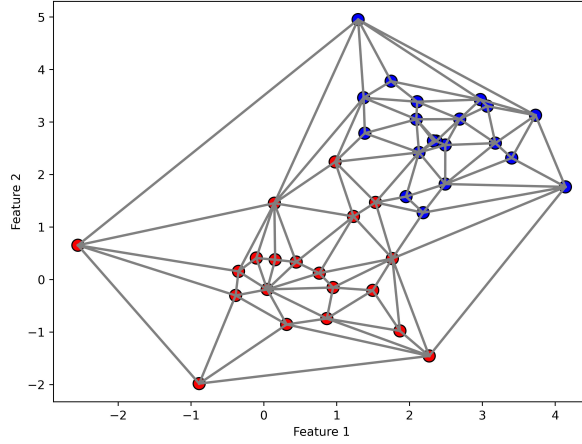


Figure 7: Toy Dataset-1 along with its Delaunay graph.

The obtained number of Graver elements is **1306**, while the count of initial feasible solutions is **349**. Figure 8 displays the sorted negated objective values. The objective function is negated to frame the maximization problem as a minimization one.

Figure 9(a) showcases the results of the augmentation procedure, plotted in sorted order of the final objective values. It provides insight into the initial objective value of each point and the number of iterations required before concluding the augmentation process. Notably, we observe that the maximum number of augmentation steps for a point does not exceed 40. Additionally, several initial points did not undergo any augmentation steps due to limitations in the truncated Graver set, which lacked sufficient directions to optimize these points.

Despite these observations, the optimum solution obtained through the augmentation process proves effective. It successfully captures the Models of Classes Distribution (MCD) points and trims down the training dataset. In Figure 9(b), subsets of the best solution are highlighted, where green points represent those belonging to the S_1 subset, effectively separating the other two subsets, each encompassing points from distinct classes. Figure 9(c) further illustrates the MCD points derived from this optimal solution, showcasing the algorithm’s commendable performance..

We executed the Ising Prediction model as described earlier, generating a 20x20 grid within the range of our original dataset to make predictions using only the Models of Classes Distribution (MCD) points. The outcomes were then utilized to create a contour plot illustrating the decision boundary. Figure 10 displays the results of the Ising Prediction model, highlighting the MCD points and the original full dataset.

Similarly, we use another synthetic 2D dataset generated using a 2x2 grid with a total of 100 data points. Randomness is introduced through shuffling the grid points and adding normal distribution perturbations to both features x_1 and x_2 coordinates. Classes were allocated in a pattern resembling a 2x2 chessboard. Figure 11 presents the dataset with its Delaunay 1-simplices graph.

The number of Graver elements obtained is **2604**, with an initial count of **123** feasible solutions. Figure 12 displays the sorted negated objective values.

Figure 13(a) presents the outcomes of the augmentation procedure applied to this dataset. It’s noteworthy that a substantial portion of the initial feasible solutions faced a lack of improving directions in the truncated Graver basis elements. Despite the increased number of Graver elements compared to the previous dataset, the heightened dataset complexity demands a greater number of truncated Graver basis elements for an effective augmentation procedure.

However, the obtained optimum solution proves satisfactory, as illustrated in Figures 13(b) and 13(c). These figures depict the robustness of the achieved result, showcasing its effectiveness despite the challenges posed by the increased dataset complexity. Figure 14 shows the result of the Ising Prediction model.

5 Conclusion

In summary, our exploration delves into the unfolding realm of quantum computing, recognizing the ongoing development of quantum annealing techniques in the current quantum computational era. Our focus centers on unveiling the

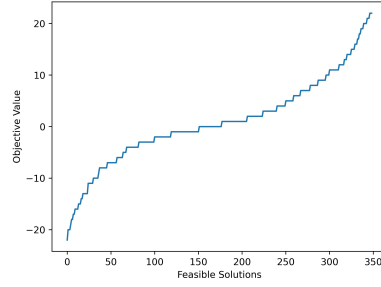
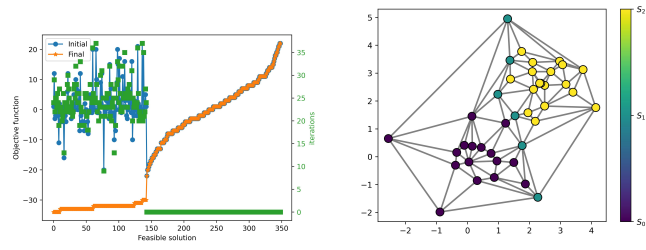
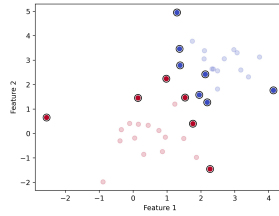


Figure 8: Sorted Objective Value of Initial Feasible Solutions for Dataset-1.



(a) Result of Augmentation (b) Optimum result showing the 3 subsets



(c) MCD points of the optimum result

Figure 9: Result of GAMA for Dataset-1.

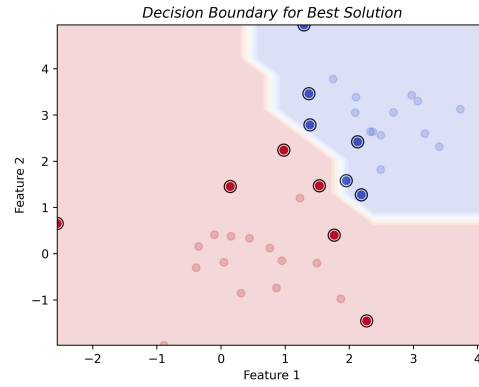


Figure 10: Decision Boundary from the Ising Prediction Model for Dataset-1

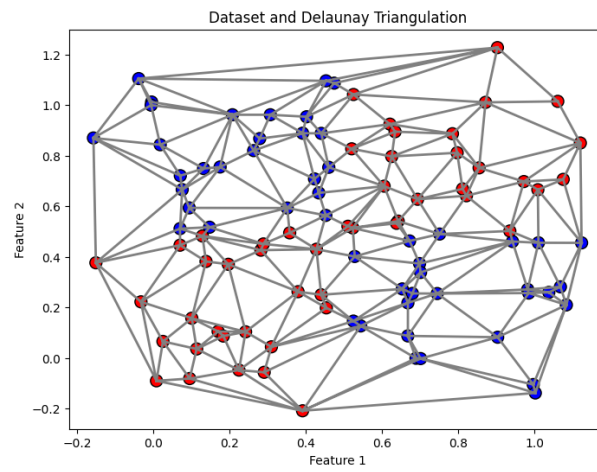


Figure 11: Dataset-2 along with its Delaunay graph.

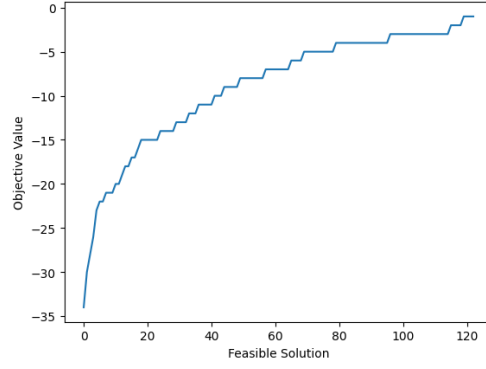
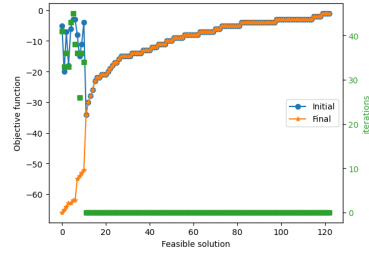
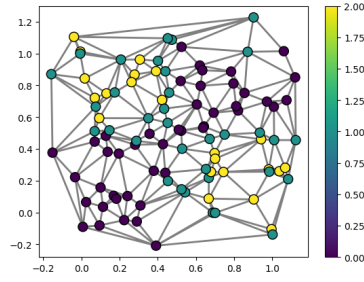


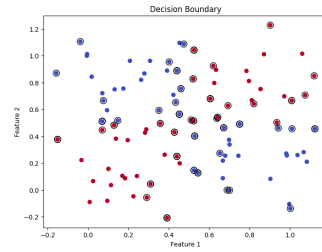
Figure 12: Sorted Objective Value of Initial Feasible Solutions for Dataset-2



(a) Result of Augmentation



(b) Optimum result showing the 3 subsets



(c) MCD points of the optimum result

Figure 13: Result of GAMA for Dataset-2.

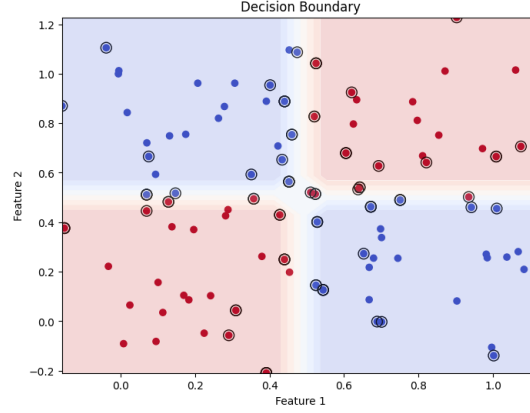


Figure 14: Decision Boundary from the Ising Prediction Model for Dataset-2

potential of GAMA within the domain of quantum machine learning, particularly for binary classification tasks.

The Ising prediction model, marked by its intrinsic probabilistic nature, emerges as a pivotal outcome of our investigation. Despite the continued evolution of quantum annealing, our findings underscore the promising performance of the Ising prediction model. Our nuanced experimentation with the β parameters unveils subtle influences on the decision boundary, suggesting a fertile ground for future exploration.

Looking forward, the adaptability and fine-tuning capabilities of the Ising prediction model present exciting prospects. The endeavor to extend this methodology to address multiple classes stands as a compelling avenue for future research. As quantum computing advances, our work contributes to the ongoing dialogue on the applications of quantum machine learning, laying the foundation for further studies that delve into the intricacies of quantum-assisted classification challenges.

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